

Paper V — The Strong Force: A Dynamical $SU(3)$ Colour Sector, Gluon Self-Coupling, and Confinement from Exact Area Law to Colour-Dielectric Condensate

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Series: “A Universe in a Bottle” — Paper V of XI. Builds on Paper I (rotation propagator) and Paper IV (gauge principle); supplies the binding for the baryon sector (Paper X).

Revision 2 (2026-06-08): re-issued for the eleven-paper series; extended beyond the exact 2D area law (Rev. 1) to the 3+1D confinement story — the colour-dielectric dual superconductor (F86), the lattice-gauge Monte-Carlo string tension (F94), and the centre-phase closure principle (F97/F98). Gluon propagator law updated to the even law forced by F91.

Abstract

We construct the strong interaction on the BCC quantum-cellular-automaton vacuum as a dynamical $SU(3)_c$ colour gauge theory built by the *same* prescription that produces electromagnetism and the weak force: a colour-octet bilinear propagated by the lattice rotation law, with Yang–Mills self-coupling supplied by the $SU(3)$ structure constants f^{abc} . The gluon octet is verified to transform in the adjoint, to conserve the colour Casimir, and to preserve link unitarity under self-coupling, all to machine precision. The gluon propagator rides the **even** rotation law — forced, not chosen, because colour coupling is chirality-blind (the F91 classification theorem). We then establish confinement at three complementary levels: (i) in two dimensions the lattice gauge theory is exactly solvable, the Wilson loop obeys a strict area law, and the static potential is exactly linear $V(R) = \sigma R$ with $\sigma(\beta) = -\ln w(\beta) > 0$ for every finite coupling; (ii) in 3+1D a colour-dielectric dual-superconductor (BPS) construction gives the flux-tube tension $\sigma_{\text{BPS}} = 2\pi v^2 n$ exactly, with n the centre charge; (iii) a 3+1D $SU(3)$ heat-bath lattice Monte-Carlo with Lüscher–Weisz multilevel sampling measures the area-law tension directly and fixes the condensate scale via $v^* = \sqrt{\sigma_A/2\pi}$. The unifying statement (the closure principle) is that a stable colour state is one whose $\mathbb{Z}_3$ centre-phase budget closes exactly (N-ality 0), and the object enforcing the budget — the colour-dielectric condensate — is itself the binder. Confinement plus the colour-singlet construction together explain why no free quark exists and why the colour-neutral baryon does — the structural prerequisite for Paper X.

1. Introduction

The strong force binds quarks into hadrons and confines colour so that no isolated quark is ever observed. In the Standard Model it is quantum chromodynamics (QCD), a non-Abelian $SU(3)$ gauge theory whose confining property, while overwhelmingly supported by lattice QCD, is not analytically proven in 3+1 dimensions (it is one of the Clay Millennium problems).

On the BCC lattice the colour sector is built by transcribing, group-for-group, the construction that produced the weak W (Paper VI) and the photon (Papers II, IV): replace the weak isospin $\tau^a/2$ by the colour generators $T^a = \lambda^a/2$ and the antisymmetric symbol ϵ^{abc} by the $SU(3)$ structure constants f^{abc} . This paper delivers (i) the dynamical gluon field with self-coupling, (ii) confinement at the exactly-solvable 2D level, and (iii) the 3+1D confinement mechanism — a colour-dielectric

condensate cross-checked against a genuine gauge-Monte-Carlo string tension — culminating in a single closure principle that the baryon sector (Paper X) inherits.

2. The dynamical gluon sector

2.1 Structure constants

The totally antisymmetric $SU(3)$ structure constants follow from $[T^a, T^b] = if^{abc}T^c$, with nine independent non-zero values

$$f^{123} = 1, \quad f^{147} = f^{246} = f^{257} = f^{345} = \frac{1}{2}, \quad f^{156} = f^{367} = -\frac{1}{2}, \quad f^{458} = f^{678} = \frac{\sqrt{3}}{2}. \quad (2.1)$$

The Jacobi identity $f^{abe}f^{ecd} + f^{bce}f^{ead} + f^{cae}f^{ebd} = 0$ holds at 1.1×10^{-16} across all 8^4 index combinations.

2.2 Colour-octet bilinear and free propagation

Following the photon/ W construction, build the colour-octet Pauli-vector bilinear from the quark colour-triplet field $q^{f,c}$ (flavour f , colour c):

$$G^{a,i}(\mathbf{x}) = \sum_f \sum_{cc'} (T^a)_{cc'} \sum_{\alpha\beta} q^{f,c,\alpha\dagger}(\mathbf{x}) \sigma_{\alpha\beta}^i q^{f,c',\beta}(\mathbf{x}), \quad a = 1, \dots, 8, \quad (2.2)$$

with **Hermitian** (not transpose) conjugation, required because $V^\top V \neq I$ for $SU(3)$. The eight components are the gluon octet, propagated by the Paper I rotation law per colour index. The octet transforms as the $SU(3)$ adjoint (residual 6.7×10^{-16} in 2D, 7.8×10^{-16} on BCC) and conserves the colour Casimir $\sum_a \|G^a\|^2$ (bit-for-bit) (Finding F43).

Propagator law — forced to be even (F91). Whereas the chiral $W^\pm$ ride the chiral law and the photon rides the even law, the gluon propagator is **even**, and this is forced rather than chosen: colour coupling acts as $g_s T^a$ on the colour index and is *blind to chirality* (both branches carry the same colour weight g_s), so the F68 commutator argument that selects the even law for the helicity-blind photon applies verbatim in colour. Accordingly the BCC gluon propagator was migrated chiral→even on 2026-06-04 (`gluon_rotation_step_spectral_bcc`; the pre-migration chiral step is retained as `..._chiral` for historical comparison). The migration changes no observable (gluons are confined) but brings the code into conformity with the forcing principle.

2.3 Yang–Mills self-coupling

The Wilson plaquette field strength is

$$G_{\mu\nu}^a(\mathbf{x}) = \frac{-i}{g a^2} \text{Tr}[T^a(U_\square - U_\square^\dagger)] = \frac{2}{g a^2} \text{Im Tr}[T^a U_\square], \quad (2.3)$$

which vanishes bit-for-bit on identity (“cold”) links because T^a is traceless. The self-coupling tick

$$\delta W^a = g \Delta t f^{abc} W^b G^c, \quad U_\ell \rightarrow e^{i\delta W^a T^a} U_\ell, \quad (2.4)$$

is the $SU(3)$ analogue of the W self-interaction and preserves link unitarity to $\sim 10^{-15}$ on both lattices (the update is a product of two $SU(3)$ elements). The quark colour current sources the gluon by the abelian-linearised Yang–Mills equation $\partial_t E^a(\mathbf{k}) = \Omega(\mathbf{k})B^a(\mathbf{k}) + gJ^a(\mathbf{k})$, with each octet component sourced only by its own current (diagonal, exact zero off-diagonal). The colour sector is thus structurally parallel to the weak sector: cold-link vacuum gives $F = 0$ exactly, constant-gauge field-strength magnitude is invariant at machine ε , and unitarity is guaranteed because $\exp(iH)U$ stays in the group.

3. Confinement I: exact area law in two dimensions

3.1 Why 2D is the rigorous testbed

In two Euclidean dimensions, after gauge-fixing to axial gauge $U_x \equiv I$, the plaquette variables are statistically independent. By Schur’s lemma the single-plaquette mean of the class-invariant Wilson distribution is

$$\langle U \rangle = w(\beta) I, \quad w(\beta) = \left\langle \frac{1}{N} \text{Re Tr } U \right\rangle, \quad d\mu_\beta(U) \propto e^{(\beta/N) \text{Re Tr } U} dU. \quad (3.1)$$

A Wilson loop enclosing area $A = R \cdot T$ is the ordered product of the enclosed independent plaquettes, so

$$\left\langle \frac{1}{N} \text{Re Tr } W(R, T) \right\rangle = w(\beta)^A \quad \text{exactly.} \quad (3.2)$$

3.2 Linear potential and string tension

Equation (3.2) gives, with no approximation,

$$\boxed{\sigma(\beta) = -\ln w(\beta) > 0 \quad \forall \beta < \infty, \quad \chi(R, T) = \sigma, \quad V(R) = \sigma R.} \quad (3.3)$$

Here $w(\beta)$ is computed deterministically by $SU(3)$ Weyl-torus quadrature (spectrally accurate, grid-converged to 3×10^{-17}). The **Creutz ratio** $\chi(R, T) = -\ln \frac{W(R, T)W(R-1, T-1)}{W(R-1, T)W(R, T-1)} = -\ln w = \sigma$ is independent of loop size to 2.2×10^{-16} , because the area difference of the four loops is exactly one plaquette. A size-independent Creutz ratio is equivalent to a strict area law, equivalent to a linear potential, equivalent to confinement. Representative string tensions:

β	0.25	0.5	1.0	2.0	4.0	8.0	20.0
$\sigma(\beta)$	4.256	3.543	2.811	2.051	1.274	0.624	0.219

σ is strictly positive at every coupling: **2D $SU(3)$ confines for all β** . This is the first place the model *predicts* confinement rather than being merely compatible with it (Finding F70).

3.3 The gradient-flow / cooling driver

A gauge-covariant Wilson gradient-flow (Lüscher) and checkerboard-cooling driver is provided and validated as a cold fixed point (bit-for-bit), action-monotone ($dS_W/dt \leq 0$), $\mathfrak{su}(3)$ -valued (5.6×10^{-16}), gauge-covariant (7.8×10^{-15}), and genuinely diffusive (acts as the lattice Laplacian, 4.9×10^{-10}). A caveat verified directly: cooling is a *smoother*, not a confiner — flowing a disordered ensemble dissolves the string tension — so the static potential must be read from the unflowed ensemble. The flow’s legitimate role is scale-setting and topological studies.

3.4 Scope of the exact result

The exact area law is special to two dimensions (independent plaquettes). Confinement in $3 + 1D$ requires either a constructive condensate model (§4) or genuine Monte-Carlo (§5); the 2D result is the rigorous, exactly-solvable anchor. The linear potential is the quenched (static-source) potential — no dynamical quark loops, hence no string breaking.

4. Confinement II: the colour-dielectric dual superconductor (3+1D)

The 3+1D confining vacuum is modelled as a **dual superconductor**: a colour-magnetic condensate that expels colour-electric flux into a thin tube (the dual Meissner effect), exactly as a type-II superconductor expels magnetic flux into Abrikosov vortices. In this picture the lattice is a **colour dielectric** $\varepsilon_c(\mathbf{x})$ — the colour analogue of the gravitational dielectric of Paper VII — whose condensate VEV v sets the confinement scale (Finding F86, “Option C”).

At the Bogomolny (BPS) point the flux-tube tension is exact:

$$\boxed{\sigma_{\text{BPS}} = 2\pi v^2 |n|} \tag{4.1}$$

with n the topological winding of the tube, equal to the source’s $\mathbb{Z}_3$ **centre charge** (triality/N-ality). The construction was built and verified (6/6) with the BPS tension exact (Tier-1). The condensate itself is obtained self-consistently from the model in Finding F88, fixing v rather than leaving it free.

The decisive identification (Finding F98) is that the centre charge n plays two roles at once: it is the **budget label** (a colour state is asymptotic iff its centre phase closes, N-ality 0) *and* the **binder coefficient** (the multiplier of the tension (4.1)). One object, two roles: the enforcer of the budget is literally the coefficient of the binding tension, so $\sigma = 0$ exactly when the budget closes and $\sigma > 0$ otherwise.

5. Confinement III: gauge Monte-Carlo string tension (3+1D)

To check that the constructive condensate matches a genuine non-Abelian gauge sector, a $3 + 1D$ $SU(3)$ heat-bath lattice Monte-Carlo with Lüscher–Weisz multilevel sampling ($84\times$ variance reduction) measures the area-law tension σ_A directly (Finding F94, “Option A”). The asymptotic k -string tensions follow the Casimir/centre law

$$\sigma_k = \frac{k(N-k)}{N-1} \sigma_1, \quad (5.1)$$

which depends only on the N-ality k (centre-periodic: $\sigma_0 = \sigma_N = 0$; adjoint screened, fundamental confined; $k \leftrightarrow N-k$ degenerate). The two 3 + 1D routes are tied together by the exact bridge $v^* = \sqrt{\sigma_A/2\pi}$: the gauge-MC string tension *is* the colour-dielectric condensate scale, re-expressed (round-trip $\sigma_C(v^*) = \sigma_A$ to $< 10^{-12}$). The two routes agree on everything the confinement claim needs — $\sigma_0 = 0$ and $\sigma_{k \neq 0} > 0$ — and differ only on the relative weight of the $k = 2$ string (Abelian-BPS $\sigma_2 = 2\sigma_1$ vs non-Abelian Casimir $\sigma_2 = \sigma_1$), a known and flagged A-vs-C gap. The model predicts the non-Abelian Casimir/sine-law regime is physical, in agreement with lattice QCD.

6. The closure principle and the road to hadrons

Confinement ($\sigma > 0$ off centre-closure) makes an isolated colour charge infinitely costly; the finite-energy configuration is the colour singlet. The unique singlet of $3 \otimes 3 \otimes 3 = 1 \oplus 8 \oplus 8 \oplus 10$ is the totally antisymmetric contraction $\varepsilon_{abc} q^a q^b q^c$, gauge-invariant because $\det V = 1$ for $SU(3)$. Stated in the closure language (Findings F97/F98):

Closure principle. A stable composite is a configuration whose phase budget closes exactly; the object that enforces the budget is itself the binding agent. For colour the budget is the $\mathbb{Z}_3$ centre phase: qqq gives $3 \times \frac{2\pi}{3} = 2\pi \equiv 0$ (closed), $q\bar{q}$ gives 0 (closed), but a diquark qq gives $\frac{4\pi}{3} \not\equiv 0$ (open, confined, never asymptotic).

This is the colour-sector instance of the same grammar that governs the pair sector (where the budget is the unitarity wrap $\pi/2$, Paper III §6 / Finding F92). A no-go theorem (F97) shows the two-body pair fixed point does not extend to three constituents — the baryon cannot be a phase-kinematic bound state — which *predicts* that baryon mass is $\sim 99\%$ field energy rather than constituent rest phase (PDG quark sum is 0.96% of m_p). The strong sector thus supplies both halves of “why no free quarks, and why a proton exists”: separating one quark from a singlet costs $\sigma R \rightarrow \infty$, while the singlet itself is colour-neutral, centre-closed, and bound. Paper X builds the proton, pion, and deuteron on this foundation.

7. Verification summary

Result	Section	Residual / status
f^{abc} Jacobi identity	§2.1	1.1×10^{-16}
Octet adjoint transform (2D / BCC)	§2.2	6.7×10^{-16} / 7.8×10^{-16}
Colour Casimir conservation	§2.2	0 (bit-for-bit)
Even gluon propagator forced (F91)	§2.2	structural (chirality-blind coupling)
Cold-link $G_{\mu\nu}^a = 0$	§2.3	0 (bit-for-bit)
Self-coupling link unitarity	§2.3	$\sim 10^{-15}$
Area law $\langle W \rangle = w^{RT}$ / Creutz ratio	§3.1–3.2	2.2×10^{-16}

Result	Section	Residual / status
Linear potential $V(R) = \sigma R$, $\sigma > 0 \forall \beta$	§3.2	$\sigma_{\min} = 0.219$
Gradient-flow gauge covariance	§3.3	7.8×10^{-15}
BPS flux-tube tension $\sigma = 2\pi v^2 n$	§4	exact (Tier-1)
A C bridge $v^* = \sqrt{\sigma_A/2\pi}$	§5	$< 10^{-12}$
$\mathbb{Z}_3$ closure ($qqq, q\bar{q} \rightarrow 0$; $qq \rightarrow 4\pi/3$)	§6	exact (integer)

Underlying findings: F43 (dynamical gluons), F70 (gradient flow + exact 2D confinement), F71 (colour-singlet baryon), F86 (colour-dielectric dual superconductor), F88 (condensate from the model), F91 (even gluon propagator), F94 (gauge-MC string tension + A C bridge), F97 (centre-phase closure no-go), F98 (enforcer = binder).

8. Discussion

The strong sector demonstrates the modularity of the construction: the *same* rotation propagator, plaquette field strength, and structure-constant self-coupling that built the weak sector build the colour sector, with only the group swapped — and the propagator law (even) is forced by the chirality-blindness of colour, not chosen. The qualitatively new physics — confinement — is delivered exactly where it can be delivered exactly (2D), modelled constructively in $3 + 1D$ as a colour dielectric, and cross-checked against a genuine gauge Monte-Carlo, with all three views unified by the centre-phase closure principle.

Two limits are explicit. First, the absolute $3 + 1D$ string tension and the QCD scale Λ remain calibration (Tier-B) items; the gauge-MC production run is user-run. Second, deriving σ as the Lagrange-multiplier price of centre-phase non-closure *inside the QCA update rule itself* (the deepest form of the closure principle) is future work. Neither affects the structural result: colour is confined, singlets are neutral and bound, and the gluon is a fully dynamical adjoint field with the correct self-coupling.

References

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5. M. Gell-Mann, “Symmetries of Baryons and Mesons,” *Phys. Rev.* **125**, 1067 (1962) — $SU(3)$ and f^{abc} .

6. Project findings: F43, F70, F71, F86, F88, F91, F94, F97, F98.

Companion papers: I (substrate), IV (electromagnetism / gauge principle), VI (weak), XI (baryons).